2 **(a)** A uniform wire AB of length 100 cm is connected between the terminals of a cell of e.m.f. 1.5 V and negligible internal resistance, as shown in Fig. 2.1.

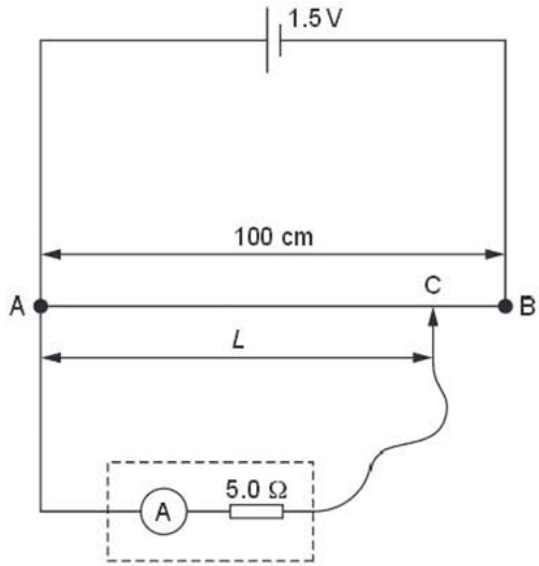

Fig. 2.1

An ammeter of internal resistance 5.0 Ω is connected to end A of the wire and to a contact C that can be moved along the wire.

Determine the reading on the ammeter for the contact C placed

(i) at A,

reading = ... A [1]

(ii) at B.

reading = ... A [1]

(b) Using the circuit in Fig. 2.1, the ammeter reading I is recorded for different distances L of the contact C from end A of the wire. Some data points are shown on Fig. 2.2.

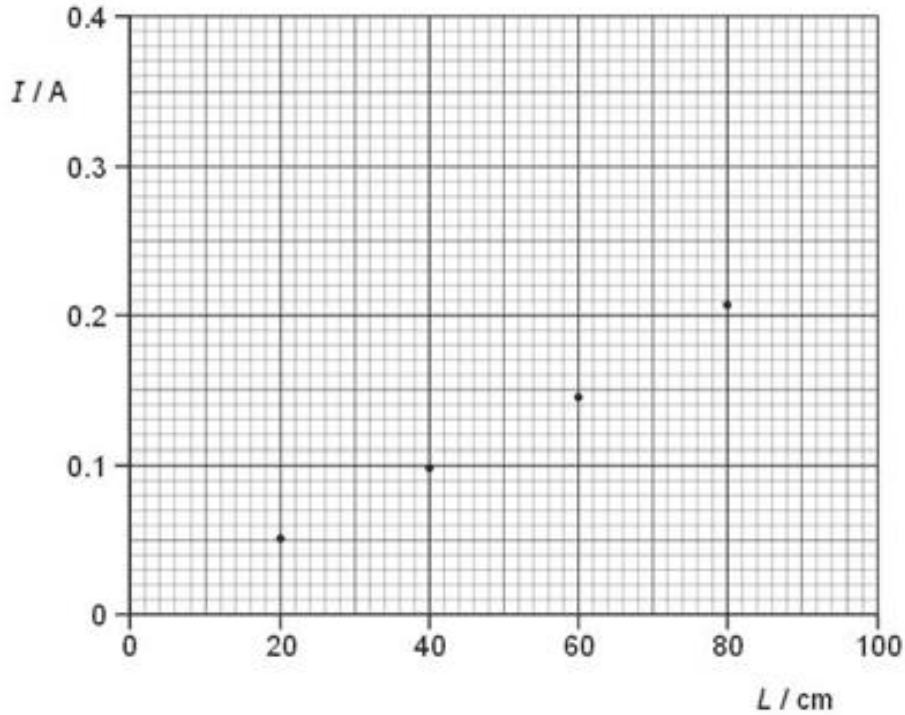

Fig 2.2

(i) Use your answers in **(a)** to plot data points on Fig. 2.2 corresponding to the contact C placed at end A and at end B of the wire. [2]

(ii) Draw a line of best fit for all of the data points and hence determine the ammeter reading for contact C placed at the midpoint of the wire.

reading = …… A [2]

(iii) Use your answer in **(ii)** to calculate the potential difference between A and the contact C for the contact placed at the midpoint of AB.

potential difference = …… V [2]

(c) Explain why, although the contact C is at the midpoint of wire AB, the answer in **(b)(iii)** is **not** numerically equal to half of the e.m.f. of the cell.

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..... [2]